JamWarden: Low-Cost Real-Time Detection of GNSS Jamming and Spoofing with Commodity Receivers

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October 2026

Abstract

Jamming and spoofing of global navigation satellite systems (GNSS) around the Baltic Sea now routinely affect aviation, shipping, and the time synchronisation of critical infrastructure. JamWarden is an open, low-cost monitor built from an ESP32 microcontroller and a u-blox receiver (about 1,000 SEK). It combines independent indicators from the receiver's RF front end (automatic gain control, noise floor, continuous-wave interference indicator), from the signal structure (C/N0 level, spread and elevation dependence) and from the navigation solution (displacement, implied speed, time offset against a local clock). These feed weighted rules with hysteresis. The same portable C++ detector runs on the microcontroller, in a PC replay tool and in a reproducible evaluation. On 280 simulated runs (70 hours), JamWarden detected all jamming of 12–25 dB and all tested spoofing attacks with median latencies of 2–19 s and no false alarms in clean, obstructed, multipath and moving-vessel conditions. A baseline using only the receivers' built-in flags detected 0–65 %. Limits are quantified: jammers below about 6 dB are detected in only 50 % of runs, and a power-matched, elevation-aware spoofer is caught only after it has displaced the position by 50 m (median 164 s).

1 Introduction

GNSS receivers are easy to disrupt because signals arrive at about -130 dBm. A jammer raises the noise floor so that satellites are lost [1]. A spoofer transmits counterfeit signals that capture the receiver's tracking loops and steer its position or time [2, 3]. Aviation authorities have issued repeated safety bulletins about GNSS outages and alterations in the region [6]. Professional interference monitoring uses dedicated hardware, but many affected users (small ports, utilities, schools, researchers) need a cheap way to know *when* and *where* their GNSS cannot be trusted.

Modern u-blox receivers expose diagnostic data such as AGC count, noise level, a CW jamming indicator and, in recent generations, a spoofing flag. These indicators are noisy and not always present, and on their own they cannot separate jamming from a blocked sky. JamWarden's contribution is a transparent detector that fuses several weak indicators with physical reasoning, together with an open hardware/firmware/dashboard stack and an evaluation protocol that reports where the approach fails.

2 Detector

Every navigation epoch (1 Hz) the detector computes:

- $\bar{C}$: mean C/N0 of the six strongest satellites; s_C: C/N0 standard deviation of satellites above 15° elevation (window mean over 10 epochs); ρ: Pearson correlation between elevation and C/N0 (window median); counts of tracked and strong (≥ 30 dB-Hz) satellites.
- From MON-RF (M9/M10) or MON-HW (M8): AGC count, noise per ms, CW jamming indicator, receiver jamming state.

- Position displacement from a frozen reference (static mode), implied speed between fixes, altitude jumps, the difference between implied and reported speed (mobile mode), and the offset between GNSS UTC and the local monotonic clock.

A 60 s baseline (mean and variance) is learned and then adapted slowly ($\alpha = 0.01$) only during clean epochs. The static reference position is never adapted, which defeats slow drag-off attacks that would otherwise move it.

Indicator	Weight	Condition (relative to baseline)
C/N0 drop	2.0	$\bar{C}_0 - \bar{C} \geq 6$ dB or no satellites
AGC anomaly	2.0	$\lvert AGC - \mu \rvert > \max(4\sigma, 0.1\mu)$
Noise rise / CW indicator	1.5 / 1.5	$> \max(4\sigma, 0.2\mu)$ / $> \mu + \max(20, 4\sigma)$
Satellites / fix lost	1.0 / 1.0	strong satellites $< 50\,\%$ of baseline / no 3D fix
Receiver jamming warning / critical	1.5 / 3.0	MON-RF/MON-HW jamming state 2 / 3
Static displacement / position jump	3.0 / 3.0	> 50 m with $h_{acc} < 25$ m / implied speed $>$ limit (latched 5 epochs)
Time jump	3.0	$\lvert\Delta\text{offset}\rvert > 1.5$ s epoch-to-epoch or vs. baseline
Receiver spoof flag	3.0	NAV-STATUS spoofDetState ≥ 2
Uniform C/N0 / no elevation dependence	1.5 / 1.5	$s_C < 2$ dB / $\rho < 0.15$ (baseline $\rho > 0.3$)
Power rise	1.0	$\bar{C} - \bar{C}_0 \geq 5$ dB

Table 1: Indicators. Jamming requires score ≥ 3.5 with at least one RF front-end indicator *and* one signal-loss indicator. Spoofing requires score ≥ 3. Alarms need 3 consecutive epochs to raise and 5 to clear.

The conjunction rule for jamming is the key to robustness: an obstruction (bridge, crane, indoor antenna) also lowers C/N0 and loses satellites, but leaves the RF front end unchanged. For spoofing, a single-antenna spoofer produces nearly equal power for all satellites and removes the natural elevation dependence of C/N0 caused by antenna gain and atmospheric path length [4]. These are cheap but effective signal-structure tests.

3 Implementation

The detector is portable C++17 without heap allocation in the parsing path (UBX state-machine parser with checksum verification, decoders, detector). It is tested with 32 unit checks, AddressSanitizer and UndefinedBehaviorSanitizer. The ESP32 firmware auto-detects the receiver baud rate, configures the needed messages through CFG-VALSET (M9/M10) or legacy CFG-MSG/CFG-PRT (M8), switches to 115,200 baud and streams JSON events and status over USB or Wi-Fi. A collector (FastAPI + SQLite) serves a live map with charts and a GeoJSON export. A physics-inspired scenario generator writes UBX streams with ground truth for 14 scenarios, including negatives and deliberately hard attacks.

4 Evaluation

Each scenario was generated with 20 random seeds (900 s each, attack onset at 240–360 s), with the receiver's own flags randomly present or absent.

Discussion. Fusing independent indicators raises detection well above the receiver's own flags without false alarms in the negative scenarios. The failures are informative. Jammers that lower C/N0 by less than the 6 dB threshold are missed in half of the runs. Lowering the threshold trades directly against false alarms from ordinary C/N0 variability. A spoofer that imitates power and elevation profiles leaves only kinematic evidence, so detection waits until the drag exceeds the 50 m radius. Antenna-array or multi-sensor techniques such as angle of arrival or cross-sensor consistency [3] are needed for faster detection. The evaluation is circular to a degree: the simulator and the detector share physical assumptions. The results therefore establish correctness and limits, not field performance.

Scenario	JamWarden	Flags only	Median / p90 latency (s)	False alarms
Broadband jammer, 12–25 dB	100 %	50 %	12 / 15	0
CW jammer	100 %	35 %	19 / 24	0
Spoofer, drag-off	100 %	45 %	8.5 / 11	0
Spoofer, 30–80 km jump	100 %	65 %	2 / 2	0
Time shift 2–10 s (with spoofer)	100 %	55 %	2 / 2	0
Time shift only	100 %	0 %	2 / 2	0
Jam then spoof (40 attacks)	100 %	45 %	10 / 13	0
Vessel, position jump (mobile mode)	100 %	50 %	2 / 2	0
Weak jammer, 4–8 dB	50 %	0 %	31.5 / 47	0
Power-matched spoofer, 0.3 m/s	100 %	45 %	163.5 / 171	0
Clean / obstruction / multipath / vessel	–	–	–	0

Table 2: Detection results over 70 simulated hours.

5 Ethics and legality

JamWarden is receive-only. Transmitting jamming or spoofing signals is illegal in Sweden and the EU, so testing relies on simulation, recorded logs, or shielded laboratories. Raw interference data with positions can reveal sensitive installations, and sharing via the GeoJSON export should be deliberate.

6 Conclusion and future work

A 1,000 SEK sensor can detect common jamming and spoofing patterns within seconds by combining RF, signal-structure, kinematic and timing evidence. Next steps are field recordings in the Baltic region, validation against public spoofing datasets processed by a software receiver [5], and multi-sensor fusion on the dashboard.

References

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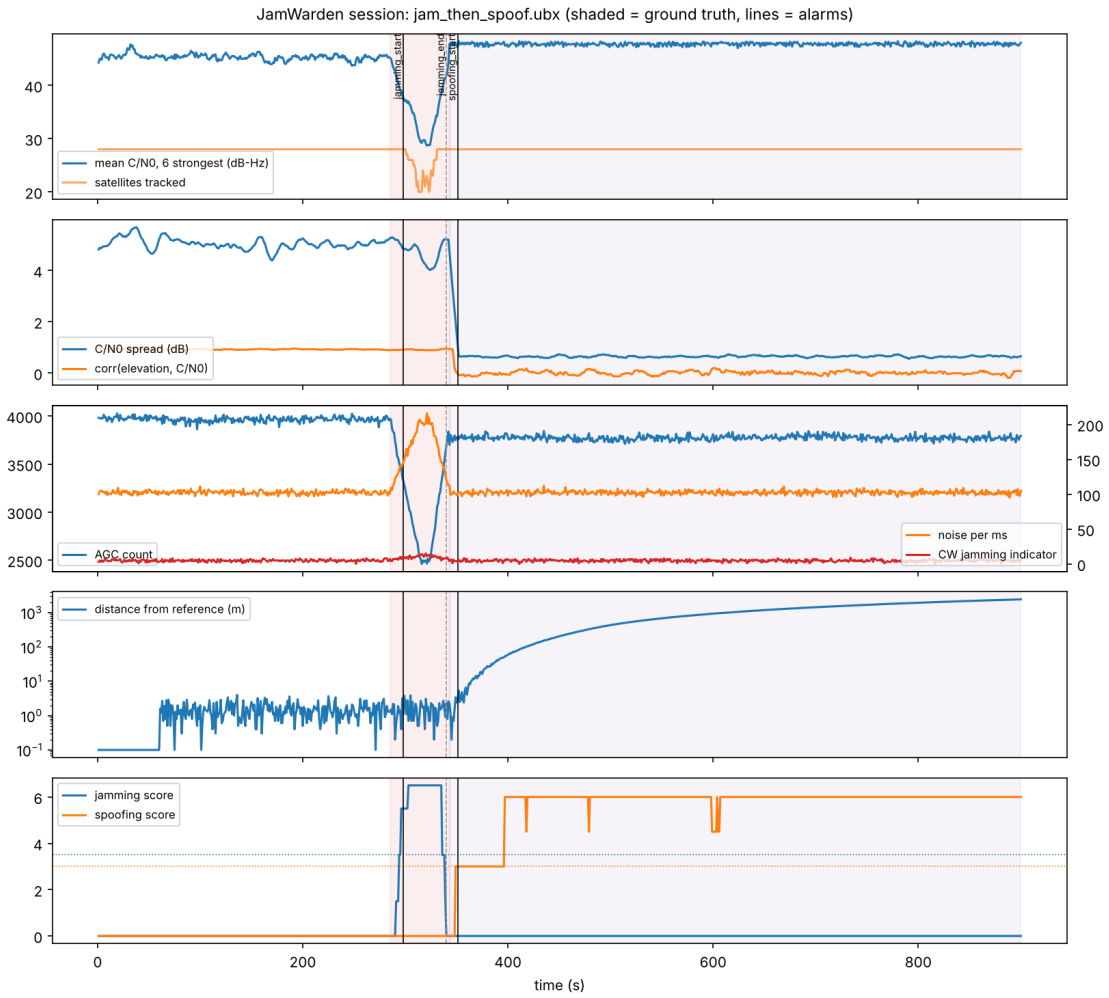


Figure 1: Jamming followed by a spoofing takeover. The C/N0 spread and elevation correlation collapse when the spoofer captures the receiver, before the position starts to move.